

Probability theory, Fall 2025

Quiz 1

October 23, 2025

Name: _____

Student No.: _____

1	10	
2	15	
3	15	
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5	15	
6	30	
Total	100	

1. (a) (5 points) If $\mathbb{P}(A)$ is 0 or 1, show that A is independent of all events B .

Proof: If $\mathbb{P}(A) = 0$, then for any event B , $A \cap B \subset A$, so $\mathbb{P}(A \cap B) = 0 = \mathbb{P}(A)\mathbb{P}(B)$.

If $\mathbb{P}(A) = 1$, then $\mathbb{P}(A^c) = 0$. $\mathbb{P}(A \cap B) = \mathbb{P}(B) - \mathbb{P}(B \cap A^c) = \mathbb{P}(B)$, so A, B are independent. $\square$

- (b) (5 points) Two fair dice are rolled. Find all positive integer values $k \in \{1, 2, \dots\}$ such that the event that the sum is k is independent of the first roll is a 4.

Solution. Let A be the event that the sum is k and B be the event that the first roll is a 4.

By (a), when $\mathbb{P}(A) = 0$, then A, B are independent. Hence, when $k = 1$ and $k \geq 13$, A, B are independent.

Note that when $k = 2, 3, 4, 11, 12$, $\mathbb{P}(A \cap B) = 0$ but $\mathbb{P}(A), \mathbb{P}(B) \neq 0$. Hence they are not independent.

When $k = 5, 6$ or $8 \leq k \leq 10$, $\mathbb{P}(A|B) = \frac{1}{6} \neq \mathbb{P}(A)$, A, B are not independent.

When $k = 7$, $\mathbb{P}(A|B) = \mathbb{P}(A) = \frac{1}{6}$, A, B are independent.

Altogether, when $k = 1, k = 7$ or $k \geq 13$, A, B are independent.

2. (15 points) An urn initially contains 2 red and 1 blue ball. At each stage, a ball is randomly chosen and then replaced along with another of the same color. Let X denote the selection number of the first chosen ball that is blue. For instance, if the first selection is red and the second blue, then X is equal to 2.

- (a) (5 points) Find $\mathbb{P}(X > k), k \geq 1$.

Solution. $\{X > k\}$ means that the first k selections are all red, then the probability is

$$\mathbb{P}(X > k) = \frac{2}{3} \cdot \frac{3}{4} \cdots \frac{k+1}{k+2} = \frac{2}{k+2}.$$

- (b) (5 points) Show that with probability 1, a blue ball is eventually chosen.

Proof: Note that $\mathbb{P}(X \leq k) = 1 - \frac{2}{k+2}$ and $\lim_{k \rightarrow \infty} \mathbb{P}(X \leq k) = 1$. Hence, $\mathbb{P}(X < \infty) = 1$. Thus, with probability 1, a blue ball is eventually chosen. $\square$

- (c) (5 points) Find $\mathbb{E}[X]$.

Solution. The probability mass function of X is $\mathbb{P}(X = k) = \mathbb{P}(X > k-1) - \mathbb{P}(X > k) = \frac{2}{(k+1)(k+2)}$. Then we have

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \frac{2k}{(k+1)(k+2)} > \sum_{k=1}^{\infty} \frac{1}{k+2} = \infty.$$

Or, directly use

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} \mathbb{P}(X > k) = \sum_{k=0}^{\infty} \frac{2}{k+2} = \infty.$$

3. (15 points) You flip a coin repeatedly, and each time this coin turns up heads with probability $\frac{3}{5}$, turns up tails with probability $\frac{2}{5}$. Let Q_n denote the probability that no run of 3 consecutive tails appears in n flips.

- (a) (10 points) Derive the recursion formula for Q_n by conditioning on when the first head occurs.

Solution. Let E_n be the event that no 3 consecutive tails appear in n flips. For $n \geq 3$, by law of total probability, since the first head should occur within 3 flips,

$$\begin{aligned} Q_n &= \mathbb{P}(E_n | \text{first head at } 1^{st} \text{ flip}) \mathbb{P}(\text{first head at } 1^{st} \text{ flip}) \\ &\quad + \mathbb{P}(E_n | \text{first head at } 2^{nd} \text{ flip}) \mathbb{P}(\text{first head at } 2^{nd} \text{ flip}) \\ &\quad + \mathbb{P}(E_n | \text{first head at } 3^{rd} \text{ flip}) \mathbb{P}(\text{first head at } 3^{rd} \text{ flip}) \\ &= \mathbb{P}(E_{n-1}) \frac{3}{5} + \mathbb{P}(E_{n-2}) \frac{3}{5} \frac{2}{5} + \mathbb{P}(E_{n-3}) \frac{3}{5} \left(\frac{2}{5}\right)^2 \\ &= \frac{3}{5} Q_{n-1} + \frac{6}{25} Q_{n-2} + \frac{12}{125} Q_{n-3}. \end{aligned}$$

- (b) (5 points) Compute Q_4 .

Solution. Use the formula in (a) and the fact that $Q_0 = Q_1 = Q_2 = 1$, we get

$$\begin{aligned} Q_3 &= \frac{3}{5} + \frac{6}{25} + \frac{12}{125} = \frac{117}{125}. \\ Q_4 &= \frac{3}{5} \frac{117}{125} + \frac{6}{25} + \frac{12}{125} = \frac{561}{625}. \end{aligned}$$

The final form is not required to be exact.

4. (15 points) There are $k+1$ coins in a box. When flipped, the i th coin will turn up heads with probability $i/k, i = 0, 1, \dots, k$. A coin is randomly selected from the box and is then flipped repeatedly. Given the first flip turns up heads, what is the conditional probability that there are exactly 2 heads in the first 3 flips?

Solution. Let H be the event that the first flip is a head. Let $C_i, i = 0, 1, \dots, k$ be the events that i th coin is selected. Let E be the event that there are exactly 2 heads in the first 3 flips.

By law of total probability for conditional probability,

$$\begin{aligned}\mathbb{P}(E|H) &= \sum_{i=0}^k \mathbb{P}(E|HC_i)\mathbb{P}(C_i|H) \\ &= \sum_{i=0}^k \binom{2}{1} \frac{i}{k} \frac{k-i}{k} \mathbb{P}(C_i|H) \\ &= 2 \sum_{i=0}^k \frac{i(k-i)}{k^2} \mathbb{P}(C_i|H).\end{aligned}$$

By Bayes's formula,

$$\mathbb{P}(C_i|H) = \frac{\mathbb{P}(H|C_i)\mathbb{P}(C_i)}{\sum_{j=0}^k \mathbb{P}(H|C_j)\mathbb{P}(C_j)} = \frac{\frac{i}{k} \frac{1}{k+1}}{\sum_{j=0}^k \frac{j}{k} \frac{1}{k+1}} = \frac{2i}{k(k+1)}.$$

Altogether, we get

$$\mathbb{P}(E|H) = \frac{4}{k^3(k+1)} \sum_{i=0}^k (ki^2 - i^3) = \frac{k-1}{3k}.$$

5. (15 points) X is a nonnegative integer-valued discrete random variable satisfying

$$\frac{\mathbb{P}(X = n + 1)}{\mathbb{P}(X = n)} = \frac{n}{3(n - 4)}.$$

Give all possible values of X . What is the distribution of X ?

Solution. Note that a valid probability mass function is nonnegative, then

$$\frac{n}{3(n - 4)} \geq 0, \text{ and } n - 4 \neq 0.$$

Thus, X takes values in $\{5, 6, 7, \dots\}$.

When $n > 5$,

$$\begin{aligned} \mathbb{P}(X = n) &= \frac{\mathbb{P}(X = n)}{\mathbb{P}(X = n - 1)} \cdots \frac{\mathbb{P}(X = 6)}{\mathbb{P}(X = 5)} \mathbb{P}(X = 5) \\ &= \frac{(n - 1) \cdots 5}{(n - 5) \cdots 1} \left(\frac{1}{3}\right)^{n-5} \mathbb{P}(X = 5) \\ &= \frac{(n - 1)!}{4!(n - 5)!} \left(\frac{1}{3}\right)^{n-5} \mathbb{P}(X = 5) \\ &= \binom{n - 1}{5 - 1} \left(\frac{1}{3}\right)^{n-5} \mathbb{P}(X = 5). \end{aligned}$$

Note that the above expression is in the form of the probability mass function for a negative binomial random variable with parameters $(5, \frac{2}{3})$

It is OK to include $X = 0$, then X follows a generalized negative binomial distribution.

6. Among n people, let X be the number of pairs of people that have the same birthday (just consider 365 days in one year).

(a) (10 points) Compute $\mathbb{E}[X]$.

Solution. Let $X_{ij} = \mathbb{1}_{\{\text{ith, jth people have the same birthday}\}}$, then $\mathbb{P}(X_{ij} = 1) = 365 \left(\frac{1}{365}\right)^2 = \frac{1}{365}$ and

$$\mathbb{E}[X_{ij}] = \mathbb{E}[X_{ij}^2] = \frac{1}{365}.$$

Hence, with $X = \sum_{i < j} X_{ij}$, we get

$$\mathbb{E}[X] = \mathbb{E} \left[\sum_{i < j} X_{ij} \right] = \binom{n}{2} \frac{1}{365}.$$

(b) (10 points) Compute $\text{Var}(X)$.

Solution. We now compute $\mathbb{E}[X^2]$. Note that for $i < j < k$, we have

$$\mathbb{E}[X_{ij}X_{jk}] = 365 \left(\frac{1}{365}\right)^3 = \frac{1}{365^2}.$$

For $i < j < k < l$, we have

$$\mathbb{E}[X_{ij}X_{kl}] = \frac{1}{365^2}.$$

Hence,

$$\begin{aligned} \mathbb{E}[X^2] &= \mathbb{E} \left[\left(\sum_{i < j} X_{ij} \right)^2 \right] \\ &= \sum_{i < j} \mathbb{E}[X_{ij}^2] + 2 \binom{\binom{n}{2}}{2} \frac{1}{365^2} \\ &= \binom{n}{2} \frac{1}{365} + \binom{n}{2} \left(\binom{n}{2} - 1 \right) \frac{1}{365^2} \end{aligned}$$

Then $\text{Var}(X) = \binom{n}{2} \frac{364}{365^2}$. (This is the same as the variance of a binomial r.v. with parameters $(\binom{n}{2}, \frac{1}{365})$. However, you can only use it if E_{ij} 's are independent)

- (c) (10 points) Use the Poisson paradigm to approximate the probability that none of them have the same birthday.

Solution. Let E_{ij} be the event that i th and j th people have the same birthday, then $\mathbb{P}(E_{ij}) = \frac{1}{365}$, which is small. Also, E_{ij} 's are weakly dependent, actually they are pairwise independent, since

$$\mathbb{P}(E_{ij}E_{jk}) = \frac{1}{365^2} \text{ for } i < j < k \text{ and } \mathbb{P}(E_{ij}E_{kl}) = \frac{1}{365^2} \text{ for } i < j < k < l.$$

We use the Poisson paradigm to approximate the occurrence of E_{ij} 's, with mean $\binom{n}{2} \frac{1}{365}$. Thus, the probability is approximately

$$\exp \left\{ - \binom{n}{2} \frac{1}{365} \right\}$$

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